

UnifiedBus™ Root Table (UBRT) Specification

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1 About this document

1.1 Terms and abbreviations

Field	Description
BIOS	Basic Input and Output System.
CNA	Compact Network Address. A shortened network address format supporting 16-bit and 24-bit representations.
DMA	Direct Memory Access.
EID	Entity Identifier. An identifier assigned to an Entity that uniquely identifies the communication object identity of that Entity within a UB domain.
Entity	The basic unit by which a device allocates its own resources. Each Entity is a communication object within a UB domain.
GUID	Global Unique Identifier. A globally unique identifier of Entity, assigned at the manufacturing stage.
HPA	Host Physical Address.
MCTP	Management Component Transport Protocol.
MMIO	Memory-Mapped Input/Output.
MSG	Message.
OEM	Original Equipment Manufacturer.
OS	Operation System.
PMU	Performance Monitoring Unit.
UB	UnifiedBus. An interconnect technology and protocol stack for SuperPoD.
UBC	UB Bus Controller. A component within a UBPU that implements the UB protocol stack and provides both software and hardware interfaces.
UBPU	UB Processing Unit. A processing unit that supports the UB protocol stack and implements device specific functions.
UBRT	UnifiedBus Root Table. The root table in ACPI to describe the UB device and configurations.
UMMU	UB Memory Management Unit. The component that translates a UB memory descriptor into the home's physical address and performs permission validation.
VM	Virtual Machine.

1.2 References

UnifiedBus™ (UB) Base Specification Revision 2.0

2 Introduction

This document describes the UnifiedBus Root Table (UBRT), which serves as the root structure of the UnifiedBus (UB) system for use with the Advanced Configuration and Power Interface (ACPI). The UBRT provides the OS with the fundamental configuration and architectural information necessary to initialize and manage the UB device.

During system boot, the OS searches for the UBRT in the ACPI tables. Once located, the OS reads the UBRT header to determine the number of sub-tables, then uses the Pointer values to access each sub-table. The OS then initializes the UB device based on the information provided in these sub-tables.

If the UBRT checksum is invalid or the header fields contain invalid values, the OS should treat the UB device as unavailable and log an appropriate error. The system should continue booting, but the UB functionality will not be available until the issue is resolved.

3 UBRT Structure

The UBRT follows the standard ACPI table format with a header followed by a collection of root bus descriptors:

Table 1 Header of the boot information table reported through ACPI

Field		Offset (Byte)	Size (Byte)	Description
Header	Signature	0	4	UBRT , including the null terminator.
	Length	4	4	Total table length.
	Revision	8	1	Version number of the table. The current version number is 1 .
	Checksum	9	1	The entire table, including the checksum field, must add to zero to be considered valid.
	OEMID	10	6	OEM ID.
	OEM Table ID	16	8	ID of a data table provided by the OEM. The OEM assigns an ID to each table.
	OEM Revision	24	4	Revision number provided by the OEM.
	Creator ID	28	4	ID of the vendor that creates a table.
	Creator Revision	32	4	Revision number provided by the creator of the table.
Count		36	4	Number of sub-tables.
Sub tables[Count]		40	16 * count	See Table 2 .

Table 2 describes the Sub tables data structure.

Table 2 Sub tables data structure

Field	Offset (Byte)	Size (Byte)	Description
Type	0	1	Sub-table type. Each type corresponds to a sub-table: • 0: ubc • 1: ummu • 2: ub_reserved_mem • 3~255: reserved
Reserved	1	7	Reserved field. The default value is 0 .
Pointer	8	8	Pointer to a sub-table, that is, the physical address of the sub-table.

4 UBRT Sub-tables

4.1 UBC Information Table

The UBC information table SHALL include the UBC register space, configuration access queue information, available host physical address (HPA), interrupt ID, and the mapping between the UBC and UMMU.

Table 3 UBC information table

Field		Offset (Byte)	Size (Byte)	Description
Header	Name	0	16	Table name, fixed as ubc , including the null terminator.
	Total Size	16	4	Total size of the table, including the header.
	Version	20	1	Version number of the table. The current version number is 1 .
	Reserved	21	3	Reserved field. The default value is 0 .
	Remaining Size	24	4	Remaining size of the table.
	Checksum	28	4	Entire table must sum to zero. Note 1: Valid data refers to the data area whose size is Total Size minus Remaining Size starting from the header of this table. Note 2: If the remaining size is less than 4 bytes, the higher bits are padded with 0s before the operation.
Local CNA Start		32	4	Start CNA available for the local UBPU.
Local CNA End		36	4	End CNA available for the local UBPU (included).
Local EID Start		40	4	Start EID available for the local UBPU.
Local EID End		44	4	End EID available for the local UBPU (included).
Feature Sets		48	1	Features enabled for the UBC: - Bit 0: MMIO token value. It is the token value security feature of the resource space. The value 0 indicates that the feature is disabled, while the value 1 indicates that the feature is enabled. - Bit 1: MCTP over UB. The value 0 indicates that the feature is disabled, while the value 1 indicates that the feature is enabled.

Field	Offset (Byte)	Size (Byte)	Description
			- Other bits: reserved.
Reserved	49	3	Reserved field. The default value is 0 .
Cluster Mode	52	2	System operating mode: - 0: standalone mode. In this mode, the OS enumerates UBPU. - 1: SuperPoD mode. In this mode, the OS cannot enumerate UBPU. - Other value: reserved.
UBC Count	54	2	Number of UBUs.
UBC Structure[UBC Count]	56	-	Details about UBUs. See Table 4 .

Table 4 describes the **UBC Structure** data structure.

Table 4 UBC Structure data structure

Field	Offset (Byte)	Size (Byte)	Description
Interrupt ID Start	0	4	Start interrupt ID.
Interrupt ID End	4	4	End interrupt ID (included).
HPA Base	8	8	Base address of the available HPA.
HPA Size	16	8	Size of the available HPA.
Memory Size Limit	24	1	Maximum address bit width that can be accessed.
DMA CCA	25	1	DMA and cache coherent attribute (CCA) for the UBC: - 0: DMA is supported, but CCA is not supported. - 1: Both DMA and CCA are supported. - Other values: DMA is not supported.
UMMU Mapping	26	2	Mapping between the UBC and UMMU. The UMMU index corresponds to UMMU Structure[UMMU Count] in the UMMU information table.
Proximity Domain	28	2	Sequence number of the NUMA domain to which the UBC belongs.
Reserved	30	2	Reserved field. The default value is 0 .
MSG Queue Base	32	8	Base address of the UBC MSG queue register.
MSG Queue Size	40	8	Size of the UBC MSG queue register.
MSG Queue Depth	48	2	Interrupt depth of the UBC MSG queue.
MSG Queue	50	2	Interrupt ID of the UBC MSG queue.

Field	Offset (Byte)	Size (Byte)	Description
Interrupt			
MSG Queue Interrupt Attribute	52	1	Interrupt attribute of the UBC MSG queue: - Bit 0: level-triggered or edge-triggered - 0: level-triggered - 1: edge-triggered - Bit 1: high level (rising edge) or low level (falling edge) - 0: high level (rising edge) - 1: low level (falling edge) - Other bits: reserved.
Reserved	53	59	Reserved field. The default value is 0 .
GUID	112	16	UBC GUID.
Vendor Info	128	256	Vendor-defined information.

4.2 UMMU Information Table

The UMMU information table SHALL include the UMMU register space and PMU information, and describe the UMMU affinity.

Table 5 UMMU information table

Field		Offset (Byte)	Size (Byte)	Description
Header	Name	0	16	Table name, fixed as ummu , including the null terminator.
	Total Size	16	4	Total size of the table, including the header.
	Version	20	1	Version number of the table. The current version number is 1 .
	Reserved	21	3	Reserved field. The default value is 0 .
	Remaining Size	24	4	Remaining size of the table.
	Checksum	28	4	Entire table must sum to zero. Note 1: Valid data refers to the data area whose size is Total Size minus Remaining Size starting from the header of this table. Note 2: If the remaining size is less than 4 bytes, the higher bits are padded with 0s before the operation.
UMMU Count		32	2	Number of UMMUs.
Reserved		34	6	Reserved field. The default value is 0 .

Field	Offset (Byte)	Size (Byte)	Description
UMMU Structure[UMMU Count]	40	-	Details about UMMUs. See Table 6 .

Table 6 describes the **UMMU Structure** data structure.

Table 6 UMMU Structure data structure

Field	Offset (Byte)	Size (Byte)	Description
Base Address	0	8	Base address of the UMMU register.
Size	8	8	Size of the UMMU register.
Reserved	16	4	Reserved field. The default value is 0 .
Proximity Domain	20	2	Sequence number of the NUMA domain to which the UMMU belongs. If the value is 0xFFFF , the sequence number is invalid.
Reserved	22	2	Reserved field. The default value is 0 .
PMU Base Address	24	8	Base address of the performance monitoring unit (PMU) register corresponding to the UMMU.
PMU Size	32	8	Size of the PMU register corresponding to the UMMU.
Reserved	40	4	Reserved field. The default value is 0 .
Min TokenID	44	4	Minimum TokenID.
Max TokenID	48	4	Maximum TokenID.
Reserved	52	26	Reserved field. The default value is 0 .
Vendor ID	78	2	Vendor ID. For details, see Appendix C.1 in the <i>UnifiedBus™ (UB) Base Specification Revision 2.0</i> .
Vendor Info	80	80	Vendor-defined information.

4.3 UB Reserved Memory Information Table

In virtualization scenarios, the host OS SHALL describe the available memory range for the UBC to the guest OS through the UB reserved memory information table. This information table is reported by the VM BIOS. The physical machine BIOS does not need to report this information table.

Table 7 UB reserved memory information table

Field		Offset (Byte)	Size (Byte)	Description
Header	Name	0	16	Table name, fixed as ub_reserved_mem , including the null terminator.

Field		Offset (Byte)	Size (Byte)	Description
	Total Size	16	4	Total size of the table, including the header.
	Version	20	1	Version number of the table. The current version number is 1 .
	Reserved	21	3	Reserved field. The default value is 0 .
	Remaining Size	24	4	Remaining size of the table.
	Checksum	28	4	Entire table must sum to zero. Note 1: Valid data refers to the data area whose size is Total Size minus Remaining Size starting from the header of this table. Note 2: If the remaining size is less than 4 bytes, the higher bits are padded with 0s before the operation.
Memory Ranges Count		32	2	Number of UB reserved memory ranges.
Reserved		34	6	Reserved field. The default value is 0 .
Memory Ranges[Memory Range Count]		40	-	Memory range description. See Table 8 .

Table 8 describes **Memory Ranges**.

Table 8 Memory Ranges definition

Field	Offset (Byte)	Size (Byte)	Description
Flags	0	1	Flag of the memory segment: - bit0: - 1 indicates that the memory is mapped in 1:1 mode. - 0 indicates that the memory cannot be mapped in 1:1 mode. - Other bits: reserved.
Reserved	1	7	Reserved field. The default value is 0 .
Memory Base	8	8	Base address of the allocated UB memory. It is 64 KB aligned.
Memory Size	16	8	Size of the allocated UB memory, in bytes. It is 64 KB aligned.